

Lecture - 13

- A moment is an expectation that involves powers of a RV. $\mu'_k = E[X^k]$ $E[X] \rightarrow$ first raw moment

$E[X^2] \rightarrow$ second raw moment

$$\therefore, E[X^k] = \frac{1}{n} \sum_{i=1}^n X_i^k$$

Now, the k^{th} central moment is

$$E[(X - E[X])^k] \text{ or } \left[\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^k \right]$$

$\rightarrow$ Mean is the 1st raw moment $\Rightarrow E[X]$

$\rightarrow$ Var. is the 2nd central moment $\Rightarrow E[(X - E[X])^2]$

$\rightarrow$ the no. of unknown params. = no. of moment eqⁿ.

eg. $X \sim \text{Exp}(\lambda)$ $f(x; \lambda) = \lambda e^{-\lambda x}, x > 0$

$$E[X] = \frac{1}{\lambda}$$

$$\bar{X} = E[X] = \frac{1}{\lambda}$$

$$\Rightarrow \left[\begin{array}{c} \hat{\lambda} = \frac{1}{\bar{X}} \\ \text{MOM} \end{array} \right]$$

eg. Gumbel dist. $\beta > 0$ $E[X] = \alpha + \beta \gamma$ $\gamma = 0.577$

$$\text{Var}(X) = \frac{\pi^2}{6} \beta^2$$

$$\bar{X} = \frac{1}{n} \sum X_i$$

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$$

$$\bar{X} = \alpha + \beta \gamma \quad \& \quad \hat{\sigma}^2 = \frac{\pi^2 \beta^2}{6} \Rightarrow \beta^2 = \frac{\hat{\sigma}^2 6}{\pi^2}$$

$$\beta_{\text{MOM}} = \frac{\hat{\sigma} \sqrt{6}}{\pi}$$

$$\alpha = \bar{X} - \beta \gamma$$

$$\alpha = \bar{X} - \gamma \frac{\hat{\sigma} \sqrt{6}}{\pi}$$

eg. $X \sim U[a, b]$ $E[X] = \frac{a+b}{2}$ $\text{Var}(X) = \frac{(b-a)^2}{12}$

$$\bar{X} = \frac{a+b}{2}$$

$$\hat{\sigma}^2 = \frac{(b-a)^2}{12}$$

$$a + b = 2\bar{x} \quad b - a = 2\sqrt{3}\hat{\sigma}$$

$$2b = 2\bar{x} + 2\sqrt{3}\hat{\sigma}$$

$$\hat{b}_{MOM} = \bar{x} + \sqrt{3}\hat{\sigma}$$

$$\hat{a}_{MOM} = \bar{x} - \sqrt{3}\hat{\sigma}$$

- $X \sim \text{Exp}(\lambda)$, $E[X] = \frac{1}{\lambda}$ & $\text{Var}(X) = \frac{1}{\lambda^2}$

- $X_1, \dots, X_n$ are indep. with density $f(x; \theta)$

$$x_1, \dots, x_n$$

$$L(\theta) = \prod_{i=1}^n f(x_i; \theta)$$

$$\text{or } L(\mu, \sigma) = \prod_{i=1}^n f(x_i; \mu, \sigma)$$

$$\hat{\theta}_{MLE} = \underset{\theta}{\text{arg max}} L(\theta)$$

eg. $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Exp}(\lambda)$ $f(x_i) = \lambda e^{-\lambda x_i}$

$$L(\lambda) = \prod_{i=1}^n \lambda e^{-\lambda x_i} \Rightarrow L(\lambda) = \lambda^n e^{-\lambda \sum_{i=1}^n x_i}$$

$$l(\lambda) = \log L(\lambda) = n \log \lambda - \lambda \sum_{i=1}^n x_i$$

$$\frac{dl}{d\lambda} = \frac{n}{\lambda} - \sum_{i=1}^n x_i = 0 \Rightarrow \frac{n}{\lambda} = \sum_{i=1}^n x_i$$

$$\lambda = \frac{n}{\sum_{i=1}^n x_i} \Rightarrow \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i \Rightarrow \frac{1}{\bar{x}}$$

$$\hat{\lambda}_{MLE} = \frac{1}{\bar{x}}$$

if the double denominator is

less than 0.

The found value is maximum.

Ex. MoM & MLE for normal.

$$X_1, \dots, X_n \sim N(\mu, \sigma^2)$$

$$E[X] = \mu$$

$$\text{Var}(X) = \sigma^2$$

$$\bar{X} = \mu$$

$$\Rightarrow \boxed{\hat{\mu}_{\text{MoM}} = \bar{X}}$$

$$\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2 = \sigma^2 \Rightarrow \boxed{\hat{\sigma}_{\text{MoM}}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2}$$

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$$L(\mu, \sigma^2) = (2\pi\sigma^2)^{-n/2} e^{-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2}$$

$$\boxed{\hat{\mu}_{\text{MLE}} = \bar{X}}$$

$$\boxed{\hat{\sigma}_{\text{MLE}}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2}$$

- some prop. of MLE $\rightarrow$ bias goes to 0, variance is limiting, consistency, asymptotic normality, invariance

- asymptotic bias $\Rightarrow$ bias $(\hat{\theta}_n) \rightarrow 0$ as $n \rightarrow \infty$
- consistency $\Rightarrow \hat{\theta}_n \xrightarrow{P} \theta$ as $n \rightarrow \infty$
- asymptotic normality $\Rightarrow$ for large n ,

$$\hat{\theta}_{\text{MLE}} \approx N\left(\theta, \frac{1}{nI(\theta)}\right)$$

- variance is limiting $\rightarrow \text{Var}(\hat{\theta}_{\text{MLE}}) \approx \frac{1}{nI(\theta)}$
- invariance, $\hat{\theta}_{\text{MLE}}$ is for θ . $g(\theta) \rightarrow g(\theta)_{\text{MLE}} = g(\hat{\theta}_{\text{MLE}})$